

YEAR 11 MATHEMATICS APPLICATIONS

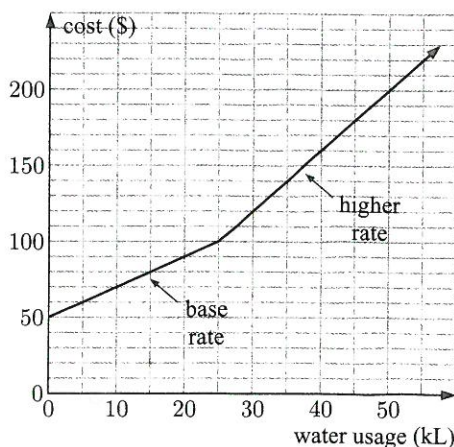
REVISION

- PIECE-WISE LINEAR GRAPHS
- STEP GRAPHS
- SIMULTANEOUS EQUATIONS
- PROBLEM SOLVING

PIECE-WISE LINEAR GRAPHS

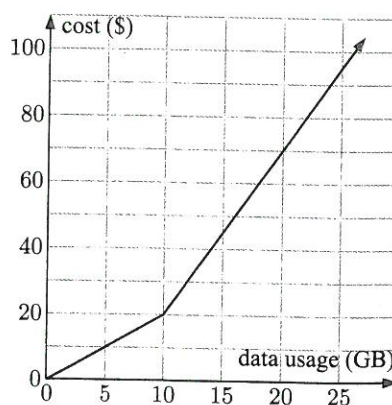
A **piece-wise linear graph** is a graph which is made up of connected straight line segments.

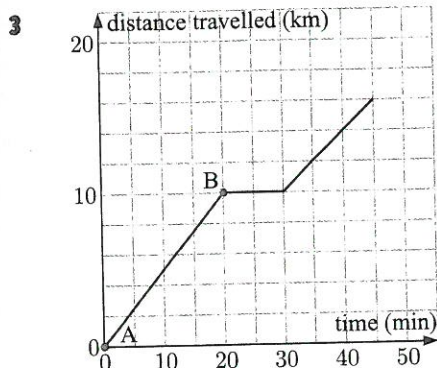
This piece-wise linear graph shows the relationship between the *water used* by a household and the *cost* of the water. Water is charged at a base rate for each litre up to 25 kL, and any water usage above 25 kL is charged at a higher rate per litre.



EXERCISE 12E.1

- 1 Consider the water costs graph on the previous page.
 - a Find the gradient of each line segment. Interpret your answers.
 - b Find the cost of using 40 kL of water.
 - c Kelly's last water bill was \$80. How much water did she use?
- 2 Each month, Colin pays \$2 per gigabyte for mobile phone data up to his allowance, and then a higher rate for excess data.
 - a What is Colin's data allowance?
 - b At what rate is Colin charged for excess data?
 - c How much does Colin need to pay for using 15 GB of data in a month?
 - d What data usage would result in a bill of \$60?





This graph shows the distance travelled by a cyclist on her way to work.

- How long did it take the cyclist to ride to work?
- How far did she travel in total?
- Find the gradient of [AB]. Interpret your answer.
- For how long did the cyclist stop to fix her tyre?
- How far had the cyclist travelled after 35 minutes?

- 4 Jasper picks berries on a farm. He earns \$5 per kg for the first 12 kg of berries he picks each day, and \$8 for each additional kilogram.

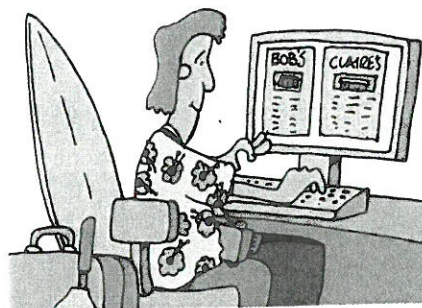
- Draw a graph to display the relationship between the *weight of berries* Jasper picks and his *daily wage*.
- How much does Jasper earn for picking 9 kg of berries?
- How many kilograms of berries must Jasper pick to earn \$100 in a day?



- 5 Shayne is looking for somewhere to stay during his vacation. He is choosing between the following places:

Bob's Beach House: \$150 per night for the first 2 nights, \$120 for each additional night
Claire's Cottage: \$180 per night for the first 3 nights, \$100 for each additional night

- On the same set of axes, graph *cost* against *number of nights* for each place.
- Find the cost of staying for 4 nights at:
 - Bob's Beach House
 - Claire's Cottage.
- For what length stay is the cost the same at each place?
- Shayne wants to stay for 8 nights. Which place will be cheaper, and by how much?



Discussion

Is it possible to write down the equation of a piece-wise linear graph?

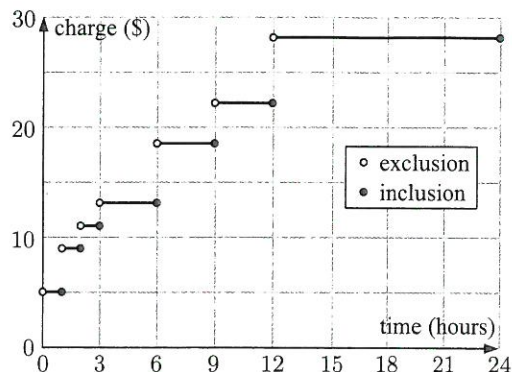
STEP GRAPHS

A **step graph** consists of a series of horizontal line segments.

For example, a step graph could be used to display the cost of parking in a car park, where the cost does not rise steadily over time, but rather in a series of discrete “jumps”.

Car Park Charges

Time	Charge
$0 < t \leq 1$ hours	\$5.00
$1 < t \leq 2$ hours	\$9.00
$2 < t \leq 3$ hours	\$11.00
$3 < t \leq 6$ hours	\$13.00
$6 < t \leq 9$ hours	\$18.00
$9 < t \leq 12$ hours	\$22.00
$12 < t \leq 24$ hours	\$28.00



Notice that:

- An open circle $\circ$ indicates that a point *is not* included.
- A filled in circle $\bullet$ indicates that a point *is* included.

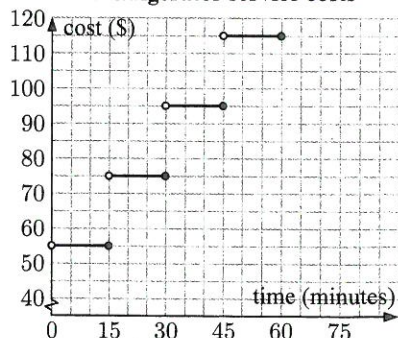
For example, parking for exactly 6 hours costs \$13, not \$18.

EXERCISE 12E.2

- Consider the step graph for car park charges above.
 - Find the cost of parking for:
 - 4 hours
 - 15 hours.
 - Friends Patrick, Sam, Ben, and George parked their cars for 9.5 hours, 8.6 hours, 12.5 hours, and 11 hours respectively. Which two of them paid the same amount for parking?
 - Jeremy paid \$9 for parking.
 - Can you determine exactly how long he parked his car for?
 - What is the maximum time he could have parked his car for?

2

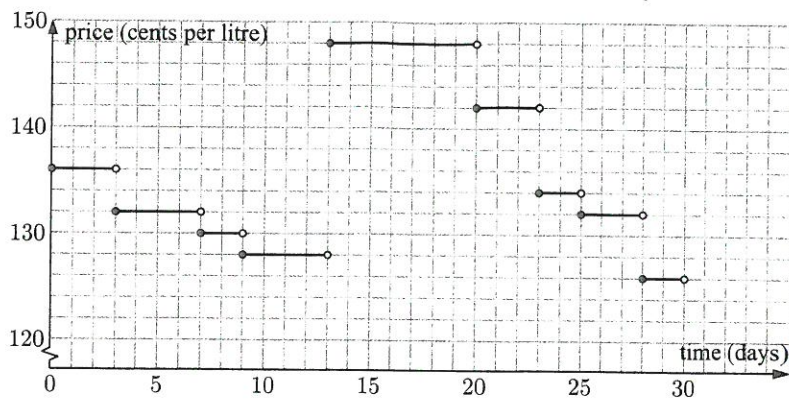
Refrigerator service costs



This step graph shows the costs for refrigerator services.

- Find the cost of a service which takes:
 - 20 minutes
 - 45 minutes.
- Find the maximum time for a service costing:
 - \$55
 - \$115
- Assuming the pattern in the graph continues, find the cost of an 80 minute service.

- 3 Rita recorded the petrol price at a service station for 30 days.



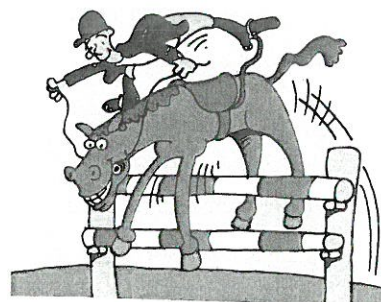
- a What was the petrol price after:
- 8 days
 - 21 days?
- b
- What was the highest petrol price?
 - How long did the price stay at this level?
- c What price occurred on two separate occasions during this period?
- d During this period, Marvin filled his car at a price of 128 cents per litre. He filled his car again 13 days later. List the possible prices he could have paid on his second visit.
- 4 In Victoria, drivers who exceed the speed limit receive demerit points. The number of demerit points depends on how much the speed limit is exceeded, as shown in the table.

Amount over speed limit (x km/h)	Demerit points
$0 < x < 10$	1
$10 \leq x < 25$	3
$25 \leq x < 35$	4
$35 \leq x < 45$	6
$x \geq 45$	8

- a Draw a step graph to display this information.
- b How many demerit points would be given to someone travelling at:
- 70 km/h in a 60 km/h zone
 - 142 km/h in a 100 km/h zone
 - 47 km/h in a 25 km/h zone?
- 5 The cost of sending a parcel overseas depends on the weight of the parcel.
- a Draw a step graph to display the information given in the table, for weights up to 5 kg.
- b Find the cost of sending a parcel weighing:
- 750 g
 - 1.6 kg
 - 3.4 kg
- c Heather has a 1.7 kg parcel and a 2.8 kg parcel to send to her mother. How much money will she save by sending the parcels together as a single package, rather than separately?

Weight	Cost
Up to 500 g	\$13
Over 500 g up to 1 kg	\$23
Over 1 kg up to 2 kg	\$32
Over 2 kg up to 3 kg	\$40
Extra kg or part thereof	\$5

- 6 In a show jumping competition, a competitor is given one-time penalty for every commenced four seconds by which they exceed the time allowed.
- For example, if the time allowed is 75 seconds, and the competitor takes 79.4 seconds, they are 4.4 seconds over the time allowed, and so are given 2 time penalties.



- a Draw a step graph to display the time penalties for different times over the time allowed.
- b If the time allowed is 82 seconds, find the time penalties given to a rider who takes:
- 83.1 seconds
 - 87.9 seconds
 - 81.5 seconds
 - 96.3 seconds

EXERCISE 12F.2

1 Use technology to solve the following simultaneous equations:

a $\begin{cases} y = 4x - 3 \\ y = -x + 5 \end{cases}$

b $\begin{cases} y = -3x - 4 \\ y = 2x + 7 \end{cases}$

c $\begin{cases} y = 2x - 3 \\ y = 10 - 6x \end{cases}$

d $\begin{cases} y = 12 - 5x \\ x + y = 7 \end{cases}$

e $\begin{cases} y = 3x - 2 \\ 7x - y = -4 \end{cases}$

f $\begin{cases} 8x - y = 16 \\ 2x + y = -10 \end{cases}$

g $\begin{cases} 5x + 4y = 6 \\ 2x + y = 5 \end{cases}$

h $\begin{cases} 3x + y = 10 \\ x - 3y = 12 \end{cases}$

i $\begin{cases} 3x - 5y = -15 \\ 8x + 3y = 24 \end{cases}$

2 Try to solve the following simultaneous equations using technology. State the number of solutions in each case.

a $\begin{cases} y = 2.5x + 6 \\ 5x - 2y = -12 \end{cases}$

b $\begin{cases} 3x + 5y = 4 \\ y = -0.6x + 2 \end{cases}$

c $\begin{cases} x + 3y = 5 \\ 4x + 12y = 20 \end{cases}$

EXERCISE 12F.3

1 Solve simultaneously:

a $\begin{cases} y = x + 2 \\ 2x + 3y = 21 \end{cases}$

b $\begin{cases} y = 2x - 3 \\ 4x - 3y = 7 \end{cases}$

c $\begin{cases} 5x + 3y = 19 \\ y = 6 - 2x \end{cases}$

d $\begin{cases} y = 3x + 1 \\ y = 7x - 1 \end{cases}$

e $\begin{cases} y = 6x - 8 \\ 3x + 2y = -6 \end{cases}$

f $\begin{cases} 4x - 7y = 1 \\ y = 11 - 3x \end{cases}$

2 Solve simultaneously:

a $\begin{cases} x = y - 3 \\ 5x - 2y = 9 \end{cases}$

b $\begin{cases} 2x - 3y = -8 \\ x = 3y - 1 \end{cases}$

c $\begin{cases} x = 4y + 3 \\ x = 9 + 7y \end{cases}$

d $\begin{cases} y = 5x - 3 \\ x = 2y + 3 \end{cases}$

e $\begin{cases} 3x + 4y = -13 \\ x = 8y - 2 \end{cases}$

f $\begin{cases} x = -5y - 2 \\ 7x + 4y = -10 \end{cases}$

3 Solve simultaneously:

a $\begin{cases} y = \frac{1}{2}x + 5 \\ 3x + 4y = 5 \end{cases}$

b $\begin{cases} x = -\frac{3}{4}y \\ 4x - 5y = -24 \end{cases}$

c $\begin{cases} x + 6y = -6 \\ y = \frac{1}{3}x - 5 \end{cases}$

d $\begin{cases} y = -\frac{1}{2}x + 3 \\ 5x + 4y = 14 \end{cases}$

e $\begin{cases} 3x + 7y = 6 \\ x = \frac{5}{3}y - 1 \end{cases}$

f $\begin{cases} 3x + 4y = 10 \\ y = \frac{3}{4}x + 2 \end{cases}$

EXERCISE 12F.4

1 Solve using the method of elimination:

a $\begin{cases} 3x - y = 5 \\ 4x + y = 9 \end{cases}$

b $\begin{cases} 5x - 2y = 17 \\ 3x + 2y = 7 \end{cases}$

c $\begin{cases} -4x + 3y = 31 \\ 4x - y = -21 \end{cases}$

d $\begin{cases} 6x + 5y = 9 \\ -6x + 7y = -45 \end{cases}$

e $\begin{cases} 2x - 3y = 18 \\ 5x + 3y = 24 \end{cases}$

f $\begin{cases} -4x + 6y = -21 \\ 4x - 2y = 11 \end{cases}$

2 Solve using the method of elimination:

a $\begin{cases} 3x + y = 16 \\ 7x - 2y = 7 \end{cases}$

b $\begin{cases} 4x + 3y = -14 \\ -x + 5y = 15 \end{cases}$

c $\begin{cases} 5x - 2y = 7 \\ 2x - y = 4 \end{cases}$

d $\begin{cases} 3x - 7y = -27 \\ -6x + 5y = 18 \end{cases}$

e $\begin{cases} 9x + 2y = -24 \\ -7x + 4y = 27 \end{cases}$

f $\begin{cases} 3x - 7y = -8 \\ 9x + 11y = 16 \end{cases}$

3 Solve using the method of elimination:

a $\begin{cases} 4x + 3y = 14 \\ 3x - 4y = 23 \end{cases}$

b $\begin{cases} 2x - 3y = 6 \\ 5x - 4y = 1 \end{cases}$

c $\begin{cases} 5x + 6y = 17 \\ 3x - 7y = 42 \end{cases}$

d $\begin{cases} 2x + 10y = -5 \\ 3x - 7y = 9 \end{cases}$

e $\begin{cases} 4x + 2y = -23 \\ 5x - 7y = -5 \end{cases}$

f $\begin{cases} 4x - 7y = 9 \\ 5x - 8y = -2 \end{cases}$

EXERCISE 12G

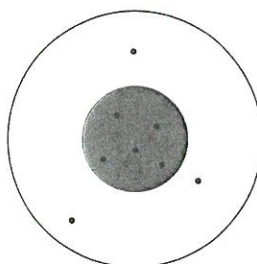
- 1 Five plates and two bowls cost \$53. Three plates and eight bowls cost \$93. Find the cost of each item.
- 2 A violinist is learning a waltz and a sonatina. One day she practices for 33 minutes by playing the waltz 4 times and the sonatina 3 times. The next day she plays the waltz 6 times and the sonatina only once, for a total of 25 minutes. Determine the length of each piece.



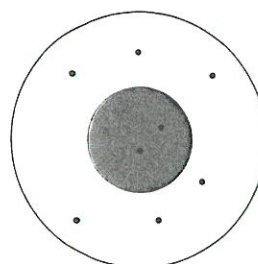
- 3 A shop sells two lengths of extension cable. Tomasz buys 2 short cables and 5 long cables with total length 26 m. Alicja buys 3 short cables and 4 long cables with total length 24.3 m. Find the two different lengths of the extension cables.

- 4 In an archery competition, competitors fire 8 arrows at a target. They are awarded points based on which region of the target is hit. The results for two of the competitors are shown opposite. How many points are awarded for hitting the:

- a red
- b blue region?



68 points

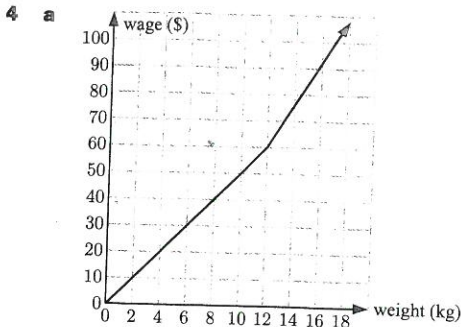


56 points

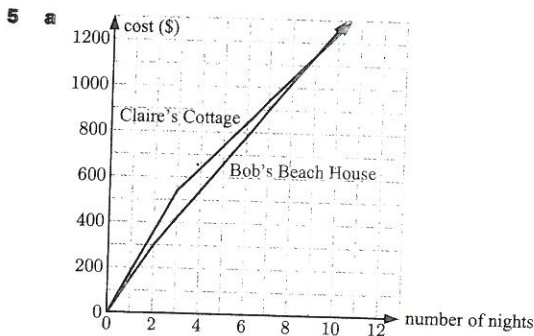
- 5 A hardware store sells 3 litre paint cans for \$15 and 5 litre paint cans for \$20. In one day the store sells 71 litres of paint worth a total of \$320. How many paint cans did the store sell?
- 6 Lidia is paid at a standard rate per hour before 5 pm, and then at a higher rate per hour after 5 pm. On Monday she worked from 2 pm to 7 pm, and earned \$110. On Tuesday she worked from 11 am to 8 pm, and earned \$195. On Wednesday Lidia worked from noon to 6 pm. How much did she earn on Wednesday?
- 7 Kristen can run at 9 km h^{-1} and walk at 3 km h^{-1} . She completed a 42 km marathon in 6 hours. What distance did Kristen run during the marathon?

EXERCISE 12E.1

- 1 a base rate: gradient = 2; the first 25 kL of water costs \$2 per kL.
higher rate: gradient = 4; each kL used above 25 kL costs \$4 per kL.
- b \$160 c 15 kL
- 2 a 10 GB b \$5 per GB c \$45 d 18 GB
- 3 a 45 min b 16 km
c gradient = 0.5; the cyclist was travelling at 0.5 km per min for the first 10 km.
d 10 min e 12 km



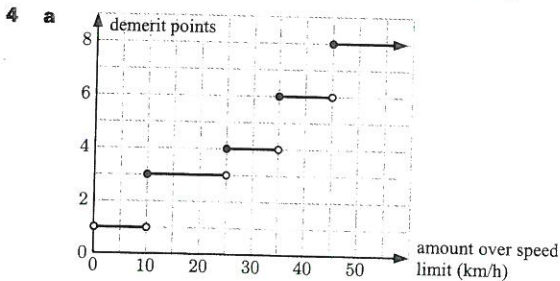
- b \$45 c 17 kg



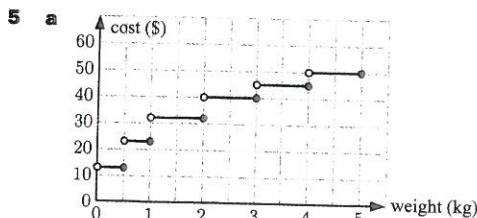
- b i \$540 ii \$640 c 9 nights
d Bob's Beach House will be cheaper by \$20.

EXERCISE 12E.2

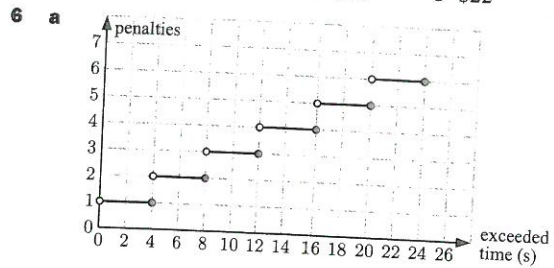
- 1 a i \$13 ii \$28 b Patrick and George
c i no ii 2 hours
- 2 a i \$75 ii \$95 b i 15 min ii 60 min
c \$155
- 3 a i 130 cents per litre ii 142 cents per litre
b i 148 cents per litre ii 7 days
c 132 cents per litre
d 142 cents per litre, 134 cents per litre, 132 cents per litre



- b i 3 points ii 6 points iii 3 points



- b i \$23 ii \$32 iii \$45 c \$22



- b i 1 time penalty ii 2 time penalties
iii 0 time penalties iv 4 time penalties

EXERCISE 12F.2

- 1 a $x = 1.6, y = 3.4$ b $x = -2.2, y = 2.6$
c $x = 1.625, y = 0.25$ d $x = 1.25, y = 5.75$
e $x = -1.5, y = -6.5$ f $x = 0.6, y = -11.2$
g $x \approx 4.67, y \approx -4.33$ h $x = 4.2, y = -2.6$
i $x \approx 1.53, y \approx 3.92$
- 2 a The lines are coincident. There are infinitely many solutions.
b The lines are parallel, so there are no solutions.
c The lines are coincident. There are infinitely many solutions.

EXERCISE 12F.3

- 1 a $x = 3, y = 5$ b $x = 1, y = -1$
c $x = -1, y = 8$ d $x = \frac{1}{2}, y = 2\frac{1}{2}$
e $x = \frac{2}{3}, y = -4$ f $x = 3\frac{3}{25}, y = 1\frac{16}{25}$
- 2 a $x = 5, y = 8$ b $x = -7, y = -2$
c $x = -5, y = -2$ d $x = \frac{1}{3}, y = -1\frac{1}{3}$
e $x = -4, y = -\frac{1}{4}$ f $x = -1\frac{11}{31}, y = -\frac{4}{31}$
- 3 a $x = -3, y = 3\frac{1}{2}$ b $x = -2\frac{1}{4}, y = 3$
c $x = 8, y = -2\frac{1}{3}$ d $x = \frac{2}{3}, y = 2\frac{2}{3}$
e $x = \frac{1}{4}, y = \frac{3}{4}$ f $x = \frac{1}{3}, y = 2\frac{1}{4}$

EXERCISE 12F.4

- 1 a $x = 2, y = 1$ b $x = 3, y = -1$
c $x = -4, y = 5$ d $x = 4, y = -3$
e $x = 6, y = -2$ f $x = 1\frac{1}{2}, y = -2\frac{1}{2}$
- 2 a $x = 3, y = 7$ b $x = -5, y = 2$
c $x = -1, y = -6$ d $x = \frac{1}{3}, y = 4$
e $x = -3, y = 1\frac{1}{2}$ f $x = \frac{1}{4}, y = 1\frac{1}{4}$
- 3 a $x = 5, y = -2$ b $x = -3, y = -4$
c $x = 7, y = -3$ d $x = 1\frac{1}{4}, y = -\frac{3}{4}$
e $x = -4\frac{1}{2}, y = -2\frac{1}{2}$ f $x = -28\frac{2}{3}, y = -17\frac{2}{3}$

EXERCISE 12G

- 1 plates: \$7, bowls: \$9
2 waltz: 3 minutes, sonatina: 7 minutes
3 short cable: 2.5 m, long cable: 4.2 m
4 a 10 points b 6 points
5 19 paint cans 6 \$125 7 36 km

ANSWERS